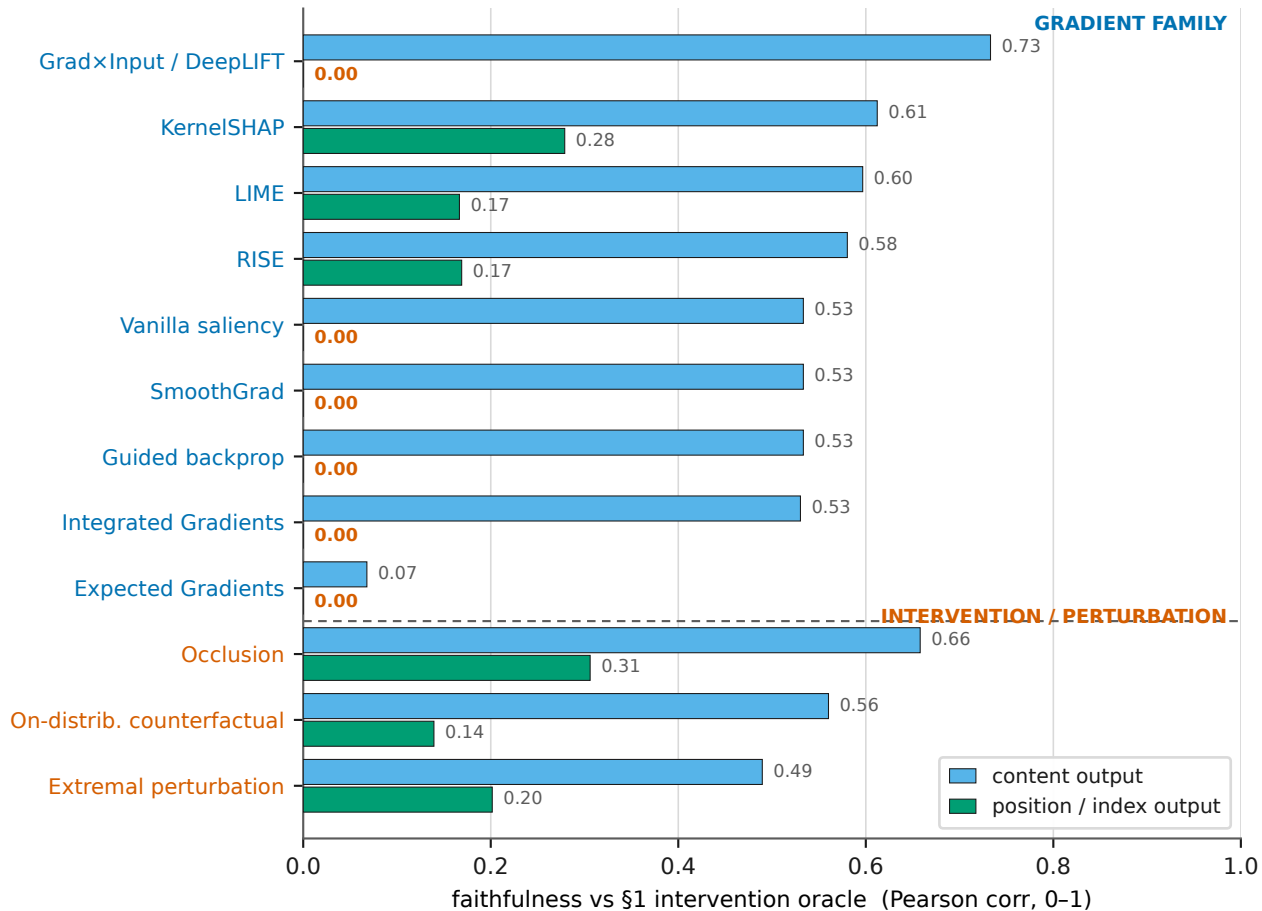


Attribution (Phase B) vs mechanistic interpretability (Phase C): faithfulness on the bit-exact VCS

On the POSITION/INDEX regime (discrete sprite-position outputs, naive gradient provably zero): the whole gradient family collapses toward chance while intervention methods hold. Causal/intervention bucket 0.412 (± 0.390 , $n=4$) vs gradient/correlational 0.068 (± 0.070 , $n=9$) — a 0.344 faithfulness gap. Activation patching = 1.000 (oracle ceiling) vs vanilla saliency = 0.000 (naive-gradient floor).

(a) Phase B — attribution, split by output regime



(b) Phase C — mechanistic interpretability

